Tic-Tac-Toe Mini Project

Course: ENGG1410 - Introductory Programming for Engineers

Project: Mini Project #1 (Part 1 & 2)

Description

This project is a C-based simulation of the classic Tic-Tac-Toe game. It supports customizable grid sizes (from 3x3 to 10x10) and features two game modes:

1. **Player vs. Player:** Two human players alternate turns.
2. **Player vs. AI:** A human player competes against a computer opponent. The AI utilizes a defense/attack strategy to block winning moves and capitalize on opportunities.

Additionally, this version includes **File I/O capabilities**, allowing users to save their game progress and resume it later.

Files Included

- TicTacToe.c: The main source code file containing the game logic, AI, and file handling.
- ticktock.dat: Binary data file used to store saved game states (created automatically).
- README: This documentation file.

Prerequisites & Compilation

This program is written in C. It includes <conio.h>, which is standard for Windows environments (MinGW/Visual Studio).

How to Compile

You can compile the code using GCC (GNU Compiler Collection) or any standard C compiler.

Using Command Prompt / Terminal:

gcc TicTacToe.c -o tictactoe

Note for Non-Windows Users (Linux/Mac):

The source code includes <conio.h>. If you are compiling on Linux or macOS, you may need to remove that include line if it causes errors, as the core logic relies primarily on standard <stdio.h> functions (scanf, getchar).

How to Run

After compiling, execute the program from the command line:

Windows:

tictactoe.exe

Linux/Mac:

./tictactoe

How to Play

Main Menu

Upon starting, you will be presented with a menu:

1. **New Game:** Start a fresh session. You will be asked to enter a grid size (3-10) and choose the game mode.
2. **Load Game:** Resume a previously saved game from ticktock.dat.
3. **Exit:** Close the application.

In-Game Controls

- **Grid Coordinates:** The board uses 1-based indexing.
- **Making a Move:** When prompted, enter the **Row Number** followed by the **Column Number** separated by a space.
    - *Example:* To place a marker in the top-left corner, enter: 1 1

Save & Quit Functionality

To implement the requirement for File I/O, the save feature is triggered during the input phase.

1. **Triggering Save:** When asked for a "Line" and "Column", enter a **negative number** (e.g., -1 -1).
2. **Confirmation:** The game will display: Do you want to save? Y/N.
3. **Saving:** Press Y (or y) to save your current board, scores, and round number. The game will save to ticktock.dat and exit.

4. **Loading:** Select **Option 2** from the main menu next time you run the program to pick up exactly where you left off.

Notes

- **AI Logic:** The computer analyzes rows, columns, and diagonals to calculate a "percentage of win." If the threat is high, it defends; otherwise, it attempts to attack.
- **Visuals:** The game uses ANSI color codes (Green for 'X', Yellow for 'O') for better visibility in supported terminals.